UC SAN DIEGO / ECE 251C

15

Wavelets and the quest for sparsity

Executive summary

An orthogonal wavelet transform changes coordinates without changing energy. We use the Haar matrices to make analysis and synthesis concrete, then study how sparsity guides the recovery of a signal from noisy observations. Counting nonzero coefficients leads to hard thresholding; replacing that count by the ℓ^1 norm leads to soft thresholding. We also clarify the conventional ℓ^0 notation by examining what actually happens as p approaches zero. These ideas connect wavelet denoising with sparse recovery and compressed sensing.

15.1 The Haar DWT for two samples

In the previous lecture, we expressed the finite DWT as multiplication by an orthogonal matrix. We now construct this matrix explicitly for the Haar transform. This will make the coefficient ordering, the normalization, and the inverse transform concrete before we use them in a denoising problem.

Start with a signal containing only two samples:

$$x = \begin{pmatrix} x[0] \\ x[1] \end{pmatrix} \in \mathbb{R}^2.$$

One level of the Haar DWT produces one approximation coefficient and one detail coefficient:

$$a[0] = \frac{x[0] + x[1]}{\sqrt{2}}, \qquad d[0] = \frac{x[0] - x[1]}{\sqrt{2}}. \tag{15.1}$$

These are the normalized average and normalized difference of the two samples. The first coefficient measures what the samples have in common, while the second measures their variation. We use the adjacent-pair, left-minus-right convention from the previous lecture.

The factor $1/\sqrt{2}$ makes the transform preserve energy:

$$|a[0]|^2 + |d[0]|^2 = \frac{(x[0] + x[1])^2 + (x[0] - x[1])^2}{2} = |x[0]|^2 + |x[1]|^2.$$

Thus $a[0]$ is a normalized average, rather than the arithmetic mean $(x[0]+x[1])/2$. Both the averaging and differencing directions have Euclidean norm one.

Writing (15.1) in matrix form gives

$$\boxed{\begin{pmatrix} a[0] \\ d[0] \end{pmatrix} = \mathrm{W}_2 x, \qquad \mathrm{W}_2 \coloneqq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.} \tag{15.2}$$

Each row specifies the weights used to compute one output coefficient. The two rows are perpendicular, since $(1)(1)+(1)(-1)=0$. Together with their unit norms, this gives

$$\mathrm{W}_2^{\mathsf{T}} \mathrm{W}_2 = \mathrm{W}_2 \mathrm{W}_2^{\mathsf{T}} = \mathrm{I}_2.$$

The matrix is therefore orthogonal, and its transpose implements the inverse transform.

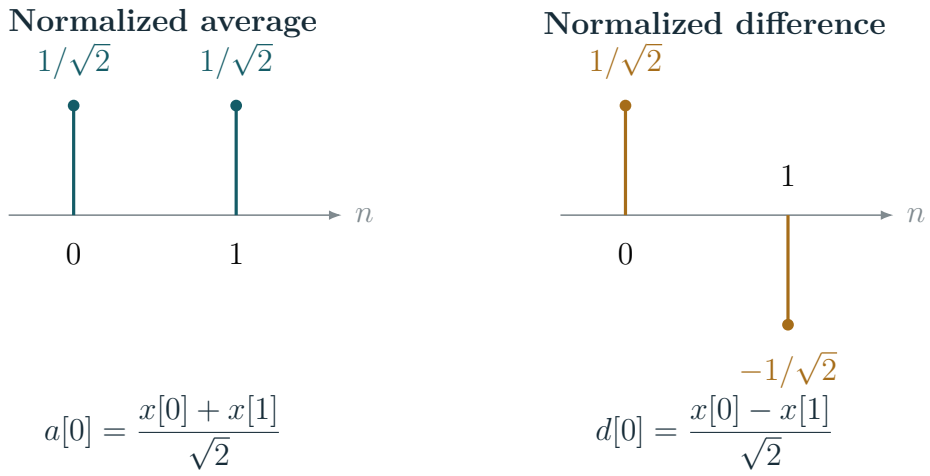


Figure 15.1: The two rows of W_2. Their inner products with the sample pair give a normalized average and a normalized difference. Both rows have Euclidean norm one; the first coefficient is $\sqrt{2}$ times the ordinary arithmetic mean.

15.2 The Haar DWT for four samples

For four samples, the first stage applies the two-sample calculation to each adjacent pair:

$$a_1[n] = \frac{x[2n]+x[2n+1]}{\sqrt{2}}, \qquad d_1[n] = \frac{x[2n]-x[2n+1]}{\sqrt{2}}, \qquad n = 0, 1.$$

The approximation vector a_1 has length two, so we can apply the same calculation once more:

$$a_0[0] = \frac{a_1[0]+a_1[1]}{\sqrt{2}}, \qquad d_0[0] = \frac{a_1[0]-a_1[1]}{\sqrt{2}}.$$

Only the approximation branch is decomposed again. The first-stage details $d_1[0]$ and $d_1[1]$ are retained.

Substituting the first-stage expressions into the second stage gives

$$a_0[0] = \frac{x[0] + x[1] + x[2] + x[3]}{2},$$
$$d_0[0] = \frac{x[0] + x[1] - x[2] - x[3]}{2}.$$

Consequently, arranging the outputs from coarse to fine produces

$$\boxed{\begin{pmatrix} a_0[0] \\ d_0[0] \\ d_1[0] \\ d_1[1] \end{pmatrix} = \mathrm{W}_4 x, \qquad \mathrm{W}_4 \coloneqq \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & -\sqrt{2} \end{pmatrix}.} \tag{15.3}$$

The first row averages over all four samples. The second compares the two halves. The last two compare the samples within each half. Every row has norm one, and different rows are orthogonal. Hence

$$\mathrm{W}_4^{\mathsf{T}}\mathrm{W}_4 = \mathrm{W}_4\mathrm{W}_4^{\mathsf{T}} = \mathrm{I}_4.$$

15.3 The Haar DWT for eight samples

For eight samples, repeat the same procedure. The first stage produces $a_2, d_2 \in \mathbb{R}^4$, the second decomposes a_2 into $a_1, d_1 \in \mathbb{R}^2$, and the third decomposes a_1 into $a_0, d_0 \in \mathbb{R}$. The final coefficient order is

$$\mathrm{W}_8 x = \big(a_0[0],\, d_0[0],\, d_1[0],\, d_1[1],\, d_2[0],\, d_2[1],\, d_2[2],\, d_2[3]\big)^{\mathsf{T}}.$$

Expanding these averages and differences yields

$$\boxed{\mathrm{W}_8 \coloneqq \frac{1}{\sqrt{8}}\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} \\ 2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 \end{pmatrix}.} \tag{15.4}$$

The normalization changes with the support size. A row supported on eight samples has nonzero entries of magnitude $1/\sqrt{8}$; one supported on four has

magnitude $1/2$; one supported on two has magnitude $1/\sqrt{2}$. In every case, the sum of the squared entries is one.

The detail rows have equally many positive and negative entries. Rows at the same scale have disjoint supports. When the support of a finer row lies within that of a coarser row, the coarser row is constant on the finer support, so the positive and negative contributions cancel in their inner product. This also makes every detail row orthogonal to the constant first row. Thus

$$\mathrm{W}_8^{\mathsf{T}}\mathrm{W}_8 = \mathrm{W}_8\mathrm{W}_8^{\mathsf{T}} = \mathrm{I}_8.$$

Coefficient	Sample index n 0	1	2	3	4	5	6	7	Row factor
$a_0[0]$	+1	+1	+1	+1	+1	+1	+1	+1	$1/\sqrt{8}$
$d_0[0]$	+1	+1	+1	+1	−1	−1	−1	−1	$1/\sqrt{8}$
$d_1[0]$	+1	+1	−1	−1	0	0	0	0	$1/2$
$d_1[1]$	0	0	0	0	+1	+1	−1	−1	$1/2$
$d_2[0]$	+1	−1	0	0	0	0	0	0	$1/\sqrt{2}$
$d_2[1]$	0	0	+1	−1	0	0	0	0	$1/\sqrt{2}$
$d_2[2]$	0	0	0	0	+1	−1	0	0	$1/\sqrt{2}$
$d_2[3]$	0	0	0	0	0	0	+1	−1	$1/\sqrt{2}$

Figure 15.2: The eight row patterns of W_8, in coefficient order. Multiply every entry in a row by the factor at the right to obtain its exact weight. The colors distinguish positive and negative signs, not magnitudes. The coarsest two rows use all eight samples; each finer detail compares the two halves of a shorter dyadic block.

The matrix W_8 computes a *three-level* Haar DWT. The approximation length decreases as

$$8 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1.$$

There are four details from the first stage, two from the second, and one from the third, together with the final approximation: $4 + 2 + 1 + 1 = 8$ coefficients. More generally, for $N = 2^I$, the full transform has I levels. This is the deepest decomposition possible before the approximation consists of a single coefficient. A recurrence for the general Haar DWT matrix will be developed in the homework.

15.4 Analysis, synthesis, and energy

The matrix viewpoint extends beyond Haar. A finite periodic DWT constructed from real orthogonal wavelet filters, with the corresponding periodic boundary convention, is represented by an orthogonal matrix

$$\mathrm{W} \in \mathbb{R}^{N\times N}, \qquad \mathrm{W}^{\mathsf{T}}\mathrm{W} = \mathrm{W}\mathrm{W}^{\mathsf{T}} = \mathrm{I}_N.$$

For the full transform on $N = 2^I$ samples, write $d_i \in \mathbb{R}^{2^i}$ for the detail vector at scale i. We arrange the coefficients as

$$c \coloneqq \mathrm{W}x = \left(a_0[0],\, d_0^{\mathsf{T}},\, d_1^{\mathsf{T}}, \ldots, d_{I-1}^{\mathsf{T}}\right)^{\mathsf{T}}. \tag{15.5}$$

Thus the vector begins with $a_0[0], d_0[0], d_1[0], d_1[1]$, followed by the four entries of d_2, and so on. The total number of entries is

$$1 + \sum_{i=0}^{I-1} 2^i = 1 + (2^I - 1) = N.$$

Analysis maps samples to coefficients, and synthesis maps coefficients back to samples:

$$\boxed{c = \mathrm{W}x, \qquad x = \mathrm{W}^{\mathsf{T}}c.} \tag{15.6}$$

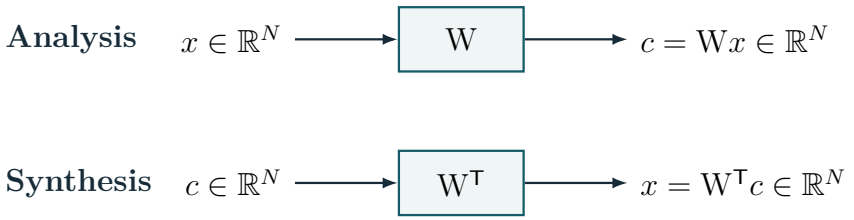


Figure 15.3: Analysis maps the N samples to N wavelet coefficients, stored from coarse to fine. Synthesis applies the transpose to recover the samples. For an orthogonal DWT, $\mathrm{W}^{\mathsf{T}}\mathrm{W} = \mathrm{I}_N$, so synthesis after analysis reconstructs the original signal.

Orthogonality also gives the energy identity

$$\|\mathrm{W}x\|_2^2 = x^{\mathsf{T}}\mathrm{W}^{\mathsf{T}}\mathrm{W}x = x^{\mathsf{T}}x = \|x\|_2^2. \tag{15.7}$$

Applying the same identity to the difference of two signals shows that Euclidean distances are preserved as well. In particular, if $\bar{c} = \mathrm{W}\bar{x}$ and $c = \mathrm{W}x$, then

$$\|\bar{x} - x\|_2 = \|\bar{c} - c\|_2. \tag{15.8}$$

Consequently, estimating the wavelet coefficients accurately is equivalent, in squared error, to estimating the signal accurately.

The purpose of the matrix representation is to expose these linear-algebraic properties. In an implementation, we still use the filter bank. For filters of fixed length, the work over all levels is proportional to

$$N + \frac{N}{2} + \frac{N}{4} + \cdots + 2 < 2N.$$

The full transform therefore costs $O(N)$ operations, or $O(LN)$ when the filter length L is included explicitly. Generic dense multiplication by an $N \times N$ matrix costs $O(N^2)$ operations. The filter-bank algorithm exploits the structure that the matrix notation summarizes.

Applying analysis followed immediately by synthesis returns the original signal. To change the signal, we can modify its coefficients between these two operations. Thresholding is one such nonlinear modification. We now derive it from an estimation problem.

15.5 Thresholding and the quest for sparsity

Consider the denoising model

$$y = x + \varepsilon, \qquad x, y, \varepsilon \in \mathbb{R}^N, \qquad \varepsilon \sim \mathcal{N}(0, \mathrm{I}_N). \tag{15.9}$$

The vector y contains the observations, x is the unknown signal, and ε is additive standard Gaussian noise. Suppose we have reason to believe that x has a sparse wavelet representation. This is an assumption about its wavelet coefficients; the samples of x themselves need not be sparse.

Applying W to the observation equation gives

$$\mathrm{W}y = \mathrm{W}x + \mathrm{W}\varepsilon.$$

Introduce the coefficient-domain quantities

$$\boxed{b := \mathrm{W}y, \qquad c := \mathrm{W}x, \qquad \xi := \mathrm{W}\varepsilon.} \tag{15.10}$$

The transformed model is then

$$b = c + \xi. \tag{15.11}$$

The noise remains standard Gaussian. To see this, a linear transformation of a Gaussian vector is Gaussian, with transformed mean and covariance. Here

$$\mathbb{E}[\xi] = 0, \qquad \mathrm{Cov}(\xi) = \mathrm{W}\mathrm{I}_N\mathrm{W}^{\mathsf{T}} = \mathrm{I}_N,$$

and therefore

$$\xi \sim \mathcal{N}(0, \mathrm{I}_N).$$

In particular, its entries remain independent standard normal random variables. The Gaussian assumption matters for this independence statement; covariance preservation alone would not establish it for arbitrary noise.

The transform has not removed noise. It has put the problem into coordinates in which the signal is assumed to be sparse, while preserving the noise distribution. In these coordinates, we seek an estimate that remains close to b but has relatively few nonzero entries.

Introduce a penalty R that is small for sparse vectors and large for vectors with many nonzero entries. We then consider

$$\boxed{\bar{c} \in \arg\min_{c \in \mathbb{R}^N} \left\{ \frac{1}{2}\|b - c\|_2^2 + \lambda R(c) \right\}, \qquad \lambda > 0.} \tag{15.12}$$

The first term measures agreement with the observed coefficients. The second expresses our preference for sparsity. The regularization parameter λ determines how strongly this preference is weighed against agreement with the noisy data. Once $\bar{c}$ is obtained, synthesis gives the signal estimate $\bar{x} = \mathrm{W}^{\mathsf{T}}\bar{c}$.

The factor $1/2$ is a convenient normalization of the squared-error term. Compared with a formulation without this factor, it rescales the parameter multiplying the penalty. We keep this convention throughout, since it determines the precise thresholds used below.

15.6 Counting nonzero coefficients

The most direct sparsity penalty counts the nonzero entries. Define

$$\operatorname{supp}(c) \coloneqq \{n \in \{0, \dots, N-1\} : c[n] \neq 0\}.$$

Its cardinality is conventionally denoted by

$$\boxed{\|c\|_0 \coloneqq |\operatorname{supp}(c)| = \sum_{n=0}^{N-1} \mathbf{1}_{\{c[n] \neq 0\}}.} \tag{15.13}$$

This quantity is often called the *zero norm*, although it is not a norm. For example, multiplying a nonzero vector by 2 leaves its nonzero count unchanged:

$$\|2c\|_0 = \|c\|_0 \neq 2\|c\|_0.$$

Thus the absolute homogeneity required of a norm fails.

The notation is motivated by a limiting relation. For $p > 0$, define

$$\|c\|_p \coloneqq \left(\sum_{n=0}^{N-1} |c[n]|^p \right)^{1/p}.$$

This is a norm for $p \geq 1$ and a quasi-norm for $0 < p < 1$. Raising it to the pth power gives

$$\|c\|_p^p = \sum_{n=0}^{N-1} |c[n]|^p.$$

For a fixed nonzero number t, we have $|t|^p = \exp(p \log |t|) \to 1$ as $p \downarrow 0$. For $t = 0$, the value is 0 for every $p > 0$. Since the sum has only finitely many terms,

$$\begin{aligned} \lim_{p \downarrow 0} \|c\|_p^p &= \sum_{n=0}^{N-1} \lim_{p \downarrow 0} |c[n]|^p \\ &= \sum_{n=0}^{N-1} \mathbf{1}_{\{c[n] \neq 0\}} = \|c\|_0. \end{aligned} \tag{15.14}$$

Remark 15.1 (What the zero-norm notation suppresses)**.** It is the pth power $\|c\|_p^p$, rather than $\|c\|_p$, that tends to the nonzero count. Formally carrying both the subscript and the exponent to zero would suggest a *zero norm to the zeroth power*, written $\|c\|_0^0$. This is only a mnemonic for the limiting expression, not ordinary exponentiation of the count: under the definition above, $(\|c\|_0)^0$ would equal 1 whenever $c \neq 0$.

The notation $\|c\|_0$ for the count is so prevalent in sparse approximation and signal recovery that we adopt the same convention. Its meaning is the definition (15.13). It is neither a norm nor, in general, the limit of $\|c\|_p$.

To see the distinction directly, fix the vector c and let

$$K \coloneqq |\operatorname{supp}(c)|.$$

If $K = 0$, then $c = 0$ and $\|c\|_p = 0$ for every $p > 0$. If $K = 1$, let n_0 be the index of the single nonzero entry. There is only one term in the sum, so

$$\|c\|_p = \big(|c[n_0]|^p\big)^{1/p} = |c[n_0]|$$

for every $p > 0$. This limit remembers the magnitude of the coefficient, whereas its nonzero count is always one.

Now suppose $K \geq 2$. Since the support is finite, the smallest nonzero magnitude

$$m \coloneqq \min_{n \in \operatorname{supp}(c)} |c[n]|$$

is strictly positive. Each of the K nonzero entries contributes at least m^p, and therefore

$$\|c\|_p = \left(\sum_{n \in \operatorname{supp}(c)} |c[n]|^p \right)^{1/p} \geq (K m^p)^{1/p} = m K^{1/p}.$$

Because $K > 1$, the factor $K^{1/p}$ tends to infinity as $p \downarrow 0$. Thus the actual limit is

$$\boxed{\lim_{p\downarrow 0} \|c\|_p = \begin{cases} 0, & K = 0, \\ |c[n_0]|, & \operatorname{supp}(c) = \{n_0\}, \\ +\infty, & K \geq 2. \end{cases}} \tag{15.15}$$

For example, $c = (3, 0)^{\mathsf{T}}$ has $\|c\|_p = 3$ for every $p > 0$, but $\|c\|_0 = 1$. For $c = (1, 1)^{\mathsf{T}}$,

$$\|c\|_p = 2^{1/p} \longrightarrow +\infty, \qquad \|c\|_p^p = 2 = \|c\|_0.$$

The nonzero count is obtained by taking the limit *after raising to the pth power*. Keeping or omitting this power changes the limiting behavior completely.

Using the nonzero count in (15.12) gives

$$\bar{c} \in \arg\min_{c\in\mathbb{R}^N} \left\{ \frac{1}{2}\|b - c\|_2^2 + \lambda \|c\|_0 \right\}. \tag{15.16}$$

Here the penalty applies to every wavelet coefficient, including the coarsest approximation. Expanding both terms reveals the structure of the problem:

$$\frac{1}{2}\|b - c\|_2^2 + \lambda\|c\|_0 = \sum_{n=0}^{N-1} \left\{ \frac{1}{2}(b[n] - c[n])^2 + \lambda \mathbf{1}_{\{c[n]\neq 0\}} \right\}. \tag{15.17}$$

Each summand depends on only one unknown coordinate. We can therefore minimize each summand independently and assemble the results. Indeed, each scalar term is bounded below by its own minimum, so the sum is bounded below by the sum of those minima. Choosing a minimizing value in every coordinate attains that lower bound. Conversely, if one coordinate did not minimize its summand, replacing just that coordinate would decrease the objective.

It remains to solve one scalar problem. Let z denote an observed coefficient and u a candidate estimate. Define

$$J_z(u) := \frac{1}{2}(z - u)^2 + \lambda \mathbf{1}_{\{u\neq 0\}}, \qquad u \in \mathbb{R}, \quad \lambda > 0. \tag{15.18}$$

15.7 Deriving the hard-thresholding rule

The penalty in (15.18) charges the same amount λ for every nonzero estimate, regardless of its magnitude. Keeping a coefficient incurs that charge; setting it to zero avoids the charge but introduces a squared error. The threshold will be the point at which these two costs agree.

The usual derivative test alone does not solve the problem, because J_z is discontinuous at zero. Separate that point from the nonzero branch:

$$J_z(u) = \begin{cases} \frac{1}{2}z^2, & u = 0, \\ \frac{1}{2}(z-u)^2 + \lambda, & u \neq 0. \end{cases}$$

If $z = 0$, then $J_0(0) = 0$, whereas $J_0(u) = \frac{1}{2}u^2 + \lambda > 0$ for every $u \neq 0$. Thus $u = 0$ is the unique minimizer.

Now suppose $z \neq 0$. For every nonzero u,

$$J_z(u) = \frac{1}{2}(z-u)^2 + \lambda \geq \lambda = J_z(z).$$

Equality holds only when $u = z$. Thus z is the unique best choice *among nonzero estimates*. The only remaining candidate is zero. We need to compare just two values:

Decision	Candidate	Objective value
Erase the coefficient	$u = 0$	$\frac{1}{2}z^2$
Retain the coefficient	$u = z$	λ

If $\frac{1}{2}z^2 < \lambda$, erasing the coefficient is cheaper, and every nonzero estimate has strictly larger cost. If $\frac{1}{2}z^2 > \lambda$, retaining it is cheaper, and $u = z$ is the unique minimizer. If the two costs are equal, both candidates minimize the objective. Since $\lambda > 0$, equality implies $z \neq 0$, so these are two distinct minimizers.

Where the threshold comes from

The cost of erasing a coefficient is $\frac{1}{2}z^2$. The least cost of keeping it nonzero is λ. Their crossover occurs when

$$\frac{1}{2}z^2 = \lambda \quad \Longleftrightarrow \quad |z| = \sqrt{2\lambda}.$$

Choose the less expensive option on either side of this crossover.

Including all three cases, the complete set of scalar minimizers is

$$\boxed{\arg\min_{u \in \mathbb{R}} J_z(u) = \begin{cases} \{0\}, & |z| < \sqrt{2\lambda}, \\ \{0, z\}, & |z| = \sqrt{2\lambda}, \\ \{z\}, & |z| > \sqrt{2\lambda}. \end{cases}} \tag{15.19}$$

To obtain a single-valued rule, choose zero in the equality case. This gives the hard-thresholding function

$$\boxed{\eta_\tau(z) := \begin{cases} 0, & |z| \le \tau, \\ z, & |z| > \tau, \end{cases} \qquad \tau := \sqrt{2\lambda}.} \tag{15.20}$$

Consequently,

$$\eta_{\sqrt{2\lambda}}(z) \in \arg\min_{u\in\mathbb{R}} J_z(u).$$

We use membership because at the threshold there is another minimizing choice. Increasing the regularization parameter raises the threshold according to $\tau = \sqrt{2\lambda}$.

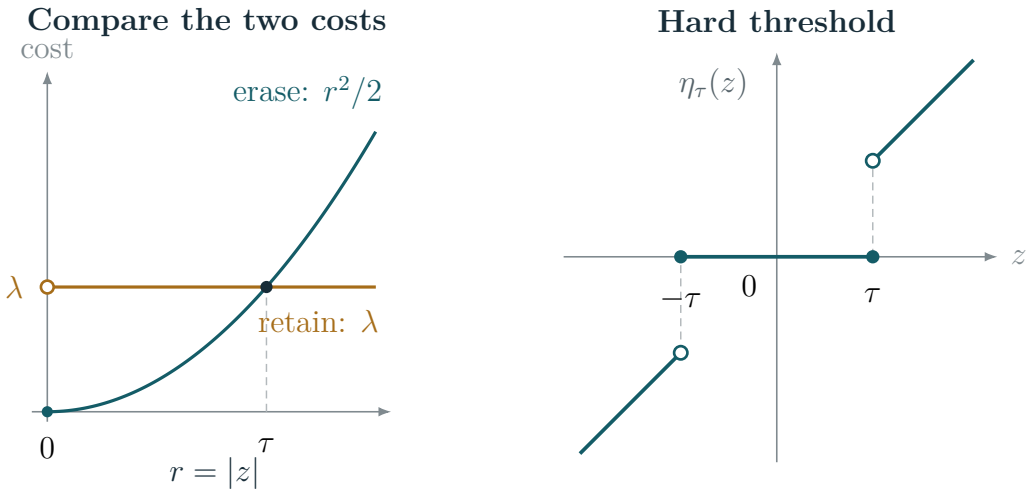


Figure 15.4: For $r = |z| > 0$, compare the erasing cost $r^2/2$ with the retaining cost λ. They meet at $\tau = \sqrt{2\lambda}$. Filled endpoints select zero at $|z| = \tau$; open endpoints exclude the identity values there. The open point at $r = 0$ excludes a nonzero retaining candidate.

15.8 Returning to the signal domain

Apply hard thresholding componentwise:

$$\big[\eta_\tau(b)\big][n] = \eta_\tau(b[n]).$$

By the separability argument, the vector

$$\bar{c} = \eta_{\sqrt{2\lambda}}(b) \in \arg\min_{c\in\mathbb{R}^N} \left\{ \frac{1}{2}\|b - c\|_2^2 + \lambda \|c\|_0 \right\} \tag{15.21}$$

is a minimizer. If some observed entries lie exactly at the threshold, the corresponding coordinates may instead be retained without changing the minimum

objective value. Our convention chooses the minimizing vector that sets all such entries to zero.

To express this estimate in signal coordinates, recall that $b = \mathrm{W}y$ and $c = \mathrm{W}x$. Orthogonality gives

$$\|b - c\|_2^2 = \|\mathrm{W}(y - x)\|_2^2 = \|y - x\|_2^2.$$

Moreover, $c = \mathrm{W}x$ is a bijective change of variables, with inverse $x = \mathrm{W}^{\mathsf{T}}c$. Thus the coefficient-domain problem is exactly equivalent to

$$\min_{x \in \mathbb{R}^N} \left\{ \frac{1}{2}\|y - x\|_2^2 + \lambda \|\mathrm{W}x\|_0 \right\}. \tag{15.22}$$

The corresponding hard-threshold estimate is

$$\boxed{\bar{x} = \mathrm{W}^{\mathsf{T}} \eta_{\sqrt{2\lambda}}(\mathrm{W}y).} \tag{15.23}$$

The argument of the thresholding function is the transform of the *observations*, $\mathrm{W}y$. The clean signal x is unknown.

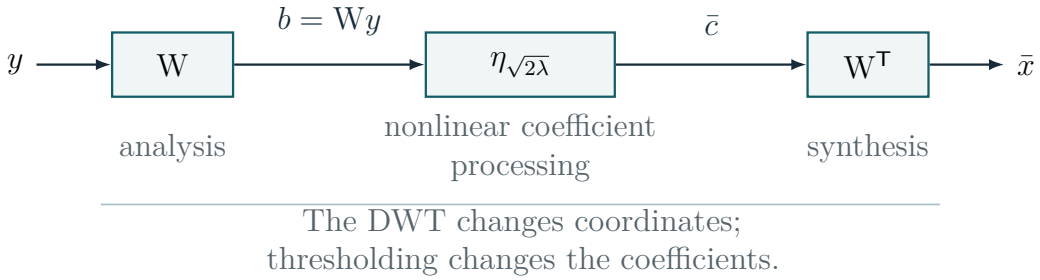


Figure 15.5: Wavelet denoising consists of analysis, nonlinear processing of the observed coefficients, and synthesis. Replacing the middle block by soft thresholding gives the corresponding ℓ^1-regularized estimate stated below.

Increasing λ raises the threshold and can only remove entries from the support of the coefficient estimate. Coefficients that survive hard thresholding keep their observed values. The method can suppress noise in coefficients that should vanish, but it can also discard small signal coefficients. The sparsity assumption motivates the rule without identifying every small observed coefficient as noise.

15.9 Beyond denoising

Ultimately, we want to recover signals from more general observations

$$y = \mathrm{H}x + \varepsilon, \qquad \mathrm{H} \in \mathbb{R}^{M \times N}, \quad y, \varepsilon \in \mathbb{R}^M.$$

The corresponding sparse recovery problem is

$$\min_{x\in\mathbb{R}^N}\left\{\frac{1}{2}\|y-\mathrm{H}x\|_2^2+\lambda\|\mathrm{W}x\|_0\right\}. \tag{15.24}$$

The same change of variables $c=\mathrm{W}x$ gives

$$\min_{c\in\mathbb{R}^N}\left\{\frac{1}{2}\|y-\mathrm{H}\mathrm{W}^{\mathsf{T}}c\|_2^2+\lambda\|c\|_0\right\}.$$

For a general acquisition matrix H, the fidelity term couples the coefficient variables. Minimizing one coordinate can change the residual associated with other coordinates, so the problem no longer separates into the scalar problems solved above. The direct denoising formula therefore does not extend to an arbitrary acquisition matrix; more general optimization methods are needed.

There is also a difficulty with the penalty itself: the nonzero count is nonconvex. This is already visible in one dimension. If $f(u)\coloneqq \mathbf{1}_{\{u\neq 0\}}$, then

$$f\left(\frac{0+1}{2}\right)=1>\frac{f(0)+f(1)}{2}=\frac{1}{2},$$

which violates convexity. Separability made the denoising problem solvable despite this nonconvexity. Once the coordinates are coupled, the same simple argument is no longer available.

Remark 15.2 (The computational difficulty of sparse recovery)**.** For a general acquisition matrix, finding a global minimizer of the ℓ^0-regularized least-squares problem is NP-hard.[1] The difficulty includes deciding which coefficients should be nonzero. Proposed supports lead to least-squares subproblems, but the general problem must also determine which support gives the best balance between fidelity and sparsity.

NP-hardness is a statement about the general class of problems: an algorithm guaranteed to solve every instance in polynomial time would imply P = NP. It does not say that every instance is difficult. Our orthogonal denoising problem is a structured case in which the objective separates and hard thresholding gives a global solution immediately. The nonconvexity of the count alone did not prevent that calculation; the coupling introduced by a general acquisition system removes the simplification.

[1] See X. Chen, D. Ge, Z. Wang, and Y. Ye, *Complexity of unconstrained L_2-L_p minimization*, *Mathematical Programming* 143 (2014), 371–383, Theorem 5, with $p=0$ and $q=2$.

15.10 A convex replacement: the ℓ^1 norm

A natural replacement for the nonzero count is

$$\|c\|_1 \coloneqq \sum_{n=0}^{N-1} |c[n]|.$$

This penalty is convex, since each absolute-value function is convex. It still encourages small coefficients to vanish, as the soft-thresholding rule below will show. Unlike the nonzero count, it also charges more for a coefficient of larger magnitude.

On the unit box $\{c \in \mathbb{R}^N : \|c\|_\infty \le 1\}$, the function $\|c\|_1$ is the *convex envelope* of $\|c\|_0$: the largest convex function that does not exceed it on this domain. This gives a precise meaning to the description of ℓ^1 as the tightest convex relaxation of ℓ^0.

To see the scalar statement first, let $f(u) \coloneqq \mathbf{1}_{\{u \neq 0\}}$ on $[-1, 1]$. The function $|u|$ is convex and satisfies $|u| \le f(u)$ there. Conversely, if g is any convex function with $g \le f$, then for $0 \le u \le 1$,

$$g(u) \le (1-u)g(0) + ug(1) \le u.$$

For $-1 \le u \le 0$, use the convex combination of 0 and -1 to obtain $g(u) \le -u$. Thus every convex lower bound lies below $|u|$.

The vector statement follows by applying this two-point argument one coordinate at a time. Write each coordinate as

$$c[n] = (1 - |c[n]|)\, 0 + |c[n]|\, \operatorname{sgn}(c[n]).$$

Combining the coordinates expresses c as a convex combination of vectors with entries zero or the corresponding signs. Coordinate n is nonzero with total weight $|c[n]|$, so the weighted nonzero count is $\sum_n |c[n]|$. Convexity therefore places every convex lower bound of $\|\cdot\|_0$ below $\|c\|_1$. Since $\|\cdot\|_1$ is itself a convex lower bound on the unit box, this proves the claim.

The domain qualification matters. Outside the unit box, $\|c\|_1$ can exceed $\|c\|_0$, so it is not a global lower bound for the count. We nevertheless use it as a convex sparsity penalty on all of $\mathbb{R}^N$. In the general inverse problem, this gives

$$\min_{x \in \mathbb{R}^N} \left\{ \frac{1}{2}\|y - \mathrm{H}x\|_2^2 + \lambda \|\mathrm{W}x\|_1 \right\}. \tag{15.25}$$

Both terms are convex in x. This change does not generally restore separability, but it removes the nonconvex penalty.

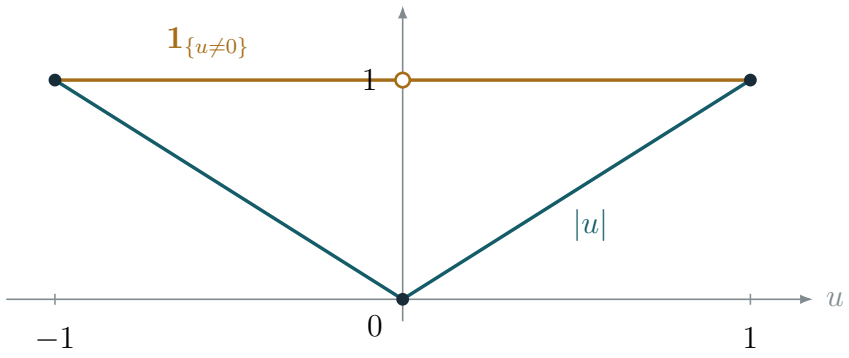


Figure 15.6: On the bounded interval $[-1, 1]$, the convex envelope of the scalar nonzero-count penalty $\mathbf{1}_{\{u\neq 0\}}$ is $|u|$. The penalty is zero at the filled origin and one at every other point of the interval; the open point at $(0, 1)$ records its jump. The bounded domain is essential to this convex-envelope statement.

Remark 15.3 (From convex relaxation to compressed sensing)**.** Replacing the nonzero count by the ℓ^1 norm is a foundational idea in compressed sensing. Suppose that $c = \mathrm{W}x$ is sparse and that we acquire $M < N$ linear measurements. In the absence of noise,

$$y = \mathrm{A}c, \qquad \mathrm{A} \coloneqq \mathrm{H}\mathrm{W}^{\mathsf{T}} \in \mathbb{R}^{M\times N}.$$

Fewer measurements than unknowns cannot determine an arbitrary vector. Sparsity restricts the possible vectors, and the acquisition matrix may still distinguish the sparse signal of interest.

The convex relaxation of finding the sparsest coefficient vector consistent with these measurements is

$$\boxed{\min_{c\in\mathbb{R}^N} \|c\|_1 \quad \text{subject to} \quad \mathrm{A}c = y.} \tag{15.26}$$

This formulation is called *basis pursuit*.[2] Under suitable conditions on the acquisition matrix and the sparsity, its solution recovers the original coefficient vector exactly, even though $M < N$. The recovery results developed by Donoho and by Candès, Romberg, and Tao in 2006 made this connection central to the theory of compressed sensing.[3]

The convex formulation makes optimization tractable; recovering the true signal still requires the stated acquisition and sparsity assumptions. With noise, exact consistency is replaced by a tolerance or a data-fidelity term, as in (15.25).

[2] S. S. Chen, D. L. Donoho, and M. A. Saunders, *Atomic decomposition by basis pursuit*, *SIAM Journal on Scientific Computing* 20 (1998), 33–61.

[3] D. L. Donoho, *Compressed sensing*, *IEEE Transactions on Information Theory* 52 (2006), 1289–1306; E. J. Candès, J. Romberg, and T. Tao, *Robust uncertainty principles: exact signal reconstruction from highly incomplete frequency information*, ibid. 52 (2006), 489–509.

15.11 Soft thresholding

We return to denoising to state the exact solution and examine how the convex penalty changes the thresholding rule.

With the ℓ^1 penalty, the coefficient-domain denoising problem is

$$\min_{c\in\mathbb{R}^N}\left\{\frac{1}{2}\|b-c\|_2^2+\lambda\|c\|_1\right\}. \tag{15.27}$$

Once again, the objective is a sum of independent scalar terms:

$$\frac{1}{2}\|b-c\|_2^2+\lambda\|c\|_1=\sum_{n=0}^{N-1}\left\{\frac{1}{2}(b[n]-c[n])^2+\lambda|c[n]|\right\}.$$

Thus we need to solve

$$\min_{u\in\mathbb{R}} F_z(u), \qquad F_z(u)\coloneqq\frac{1}{2}(z-u)^2+\lambda|u|. \tag{15.28}$$

The scalar objective is strictly convex and has a unique minimizer. That minimizer is given by the *soft-thresholding function* below; deriving the rule is left to the homework.

$$\boxed{\mathcal{S}_\lambda(z)\coloneqq\begin{cases}z+\lambda, & z<-\lambda,\\ 0, & |z|\le\lambda,\\ z-\lambda, & z>\lambda.\end{cases}} \tag{15.29}$$

Equivalently, with $(t)_+\coloneqq\max\{t,0\}$ and $\operatorname{sgn}(0)\coloneqq 0$,

$$\boxed{\mathcal{S}_\lambda(z)=\operatorname{sgn}(z)(|z|-\lambda)_+.} \tag{15.30}$$

Soft thresholding first reduces the magnitude by λ and then clips the result at zero, preserving the sign of each surviving coefficient. Its threshold is λ under our $1/2$ normalization of the squared-error term. At $z=\pm\lambda$, the unique minimizing value is zero. The two-valued minimizer set that occurred for the ℓ^0 penalty does not occur here.

As with hard thresholding, apply $\mathcal{S}_\lambda$ componentwise to vectors. The unique coefficient estimate is

$$\bar{c}=\mathcal{S}_\lambda(b).$$

Using $b=\mathrm{W}y$ and synthesizing gives

$$\boxed{\bar{x}=\mathrm{W}^{\mathsf{T}}\mathcal{S}_\lambda(\mathrm{W}y).} \tag{15.31}$$

This estimate is the unique minimizer of

$$\min_{x\in\mathbb{R}^N}\left\{\frac{1}{2}\|y-x\|_2^2+\lambda\|\mathrm{W}x\|_1\right\}. \tag{15.32}$$

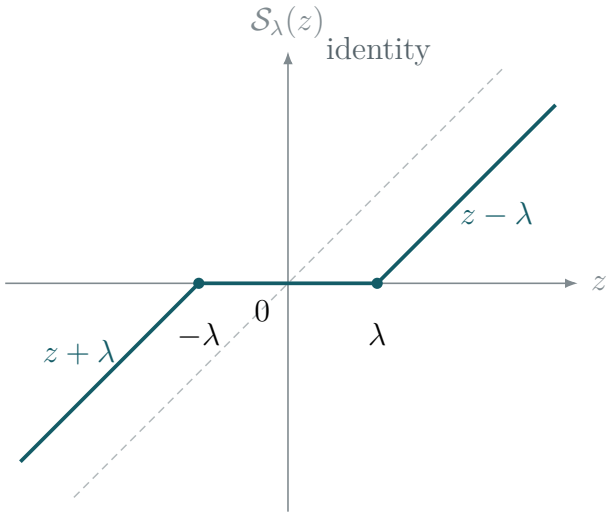


Figure 15.7: Soft thresholding is continuous. It sends the interval $[-\lambda, \lambda]$ to zero and shifts each outer identity branch toward zero by λ. The retained branches still have slope one. The dashed line is the identity map.

The change-of-variables argument is the same as for hard thresholding: the DWT preserves the data-fidelity term, and synthesis inverts the coefficient representation.

Both thresholding rules set small observed coefficients to zero. Hard thresholding leaves every retained coefficient unchanged; soft thresholding shrinks each retained coefficient toward zero. The hard rule jumps at its threshold; at equality, the underlying scalar problem has two minimizers. The soft rule is continuous and gives a unique minimizer, including at its threshold. These differences follow directly from the two penalties.

The two thresholding rules are therefore exact solutions of their respective regularized denoising problems. Orthogonality preserves the squared-error term, and separability reduces the minimization to scalar decisions. For a general acquisition system, these same penalties lead to coupled optimization problems.